

Smytten E-Commerce Database Management System

CHAPTER – 1

1. Introduction

The Smytten E-Commerce Database Management System is developed to manage and organize the information related to products, customers, trial orders, purchases, payments, and deliveries available on the Smytten platform. Smytten is a unique e-commerce platform that allows customers to explore and try sample products before purchasing full-size products. The database system stores all the required information in a structured format, making it easier to retrieve, update, and maintain records efficiently.

The system uses a relational database to establish relationships between customers, products, brands, categories, orders, and payments. It helps reduce manual work, improves data accuracy, and ensures smooth management of daily business operations.

1.1 Problem Statement

The rapid growth of online shopping has increased the need for efficient management of products, customers, trial requests, orders, and payments. In an e-commerce platform like Smytten, which offers both trial-size and full-size products, managing large volumes of data manually can create several challenges such as duplicate records, inaccurate product information, delayed order processing, inventory mismatches, and difficulty in tracking customer purchases.

As the number of users and products continues to grow, maintaining accurate and up-to-date records becomes more difficult. Without a proper database management system, it is challenging to organize customer information, manage trial products, monitor stock availability, process orders, and maintain payment records. Manual data management also increases the risk of data loss, inconsistency, and human errors, which may affect customer satisfaction and business performance.

To overcome these challenges, the Smytten E-Commerce Database Management System is developed to provide a structured and reliable solution for storing and managing data. The system organizes information into related tables, ensuring data consistency, reducing redundancy, improving security, and enabling quick retrieval of information. It also supports efficient management of products, customers, trial requests, orders, payments, and deliveries, making the overall business process faster, more accurate, and easier to manage.

1.2 Objectives

- To design a database for Smytten E-Commerce.
- To maintain customer information securely.
- To store product and trial product details efficiently.
- To manage order and payment records.
- To perform database operations such as Insert, Update, Delete, and Select.
- To reduce data redundancy and improve data consistency.
- To enable quick and accurate retrieval of information.

1.3 Motive

The motivation behind this project is to simplify the management of products available on the Smytten e-commerce platform. Since Smytten allows customers to try sample products before purchasing full-size products, an efficient database is required to manage trial requests, product details, customer information, and transactions. The database management system helps organize data efficiently, minimizes errors, improves data security, and supports faster processing of customer orders. This project also provides practical knowledge of database concepts and their application in real-world e-commerce business scenarios.

1.3 Stakeholders

The main stakeholders involved in this project are:

- **Customers** – Browse products, request trial products, place orders, and make payments.
- **Admin** – Manages products, customers, trial requests, and order records.
- **Brand Partners** – Add and update product information.
- **Delivery Partners** – Handle product delivery and update delivery status.
- **Management Team** – Uses reports for business analysis and decision-making.

1.4 Business Requirements

The database should meet the following business requirements:

- Maintain complete product and trial product information.
- Store customer details securely.
- Record customer orders and payment information.
- Update stock availability after every purchase.
- Generate accurate reports for products, trial requests, and orders.
- Support efficient data retrieval and management.
- Maintain data consistency and reduce duplication.

1.5 Functional Requirements

The system should provide the following functions:

- User registration and login.
- Add, update, delete, and view product details.
- Store customer information.
- Request trial products and purchase full-size products.
- Place and manage customer orders.
- Record payment details.
- Track order status.
- Search and retrieve product information.
- Generate reports for products and orders.

1.5 Non-Functional Requirements

Security

- User authentication through login credentials.
- Protect customer and payment information.
- Restrict unauthorized access to the database.
- Regular database backup and recovery.

Performance

- Fast data retrieval.
- Quick response to user requests.
- Efficient handling of multiple records.

Reliability

- Accurate storage and retrieval of data.
- Consistent database operations with minimal errors.

Usability

- Simple and user-friendly interface.
- Easy navigation for both customers and administrators.

Scalability

- Support future expansion by adding more products, brands, customers, trial requests, and transactions.

Availability

- The system should be available whenever users need to access product, trial, and order information.

1.6 Conclusion

The Smytten E-Commerce Database Management System provides an efficient solution for managing products, trial requests, customers, orders, payments, and deliveries. It organizes information in a structured manner, reduces data redundancy, improves security, and enables quick retrieval of records. The system also supports the unique trial-before-purchase concept of Smytten, making product management and customer transactions more efficient. This project demonstrates how a database management system can support the smooth operation of a modern e-commerce platform and improve overall business efficiency.